MUHAMMED HOGER

AI Engineer | IT & Generative AI Specialist

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PROFESSIONAL SUMMARY

Results-driven AI Engineer and IT professional with hands-on experience spanning artificial intelligence, software, and large-scale field operations in the oil & gas sector. Backed by a BSc in Information Technology and an MSc in Computer Engineering (Artificial Intelligence), I combine strong theoretical foundations with practical delivery. I specialise in generative AI, deep learning, and prompt engineering, and I am known for seeing projects through from concept to completion with measurable results. Equally effective independently and within teams, I am passionate about applying technology to solve real-world problems and streamline operations.

CORE COMPETENCIES

Generative AI: Text-to-Image Synthesis (GANs), Diffusion Models (DDRM), Vision Transformers (ViT), Image Restoration & Annotation

AI Prompting: Zero-shot, Few-shot, Chain-of-Thought (CoT), Role Prompting, Prompt Chaining, RAG (Retrieval-Augmented Generation)

Machine / Deep Learning: PyTorch, Model Training & Evaluation (IS / FID), Attention Mechanisms, Neural Network Architecture Design

Programming: Python (ML / DL), C++, SQL, HTML, CSS, JavaScript, Node.js, LaTeX

Field & Project Ops: Technical Documentation, Reporting, Data & Records Management, Cross-Team Coordination

PROFESSIONAL EXPERIENCE

Site Administrator — KNR Company (Oil & Gas) 2025 – 2026

Dana Gas to Chamchamal Pipeline Project

- Managed all site data, records, and technical documentation for a major pipeline project.
- Produced daily reports, progress logs, and compiled weekly and monthly submissions for management.
- Coordinated information flow between site teams, engineers, and project management.
- Tracked on-site personnel, equipment, and material usage to support smooth operations.

Website Administrator — Health Syndicate 2020 – 2022

Hawler, Kurdistan Region

- Created digital identities and managed work permissions for employees across the organisation.
- Maintained and supported all technical and operational website requirements.

Volunteer — Hi Technology (Hitex) 2020 – 2021

- Supported technology projects and gained hands-on experience in a professional IT environment.

EDUCATION

MSc, Computer Engineering – Artificial Intelligence 2022 – 2024

University of Kurdistan

BSc, Information Technology 2017 – 2021

Lebanese French University

RESEARCH & PROJECTS

DA²-GAN: Dual-Stage Dual-Attention GAN for Text-to-Image Synthesis MSc Thesis

- Designed a multi-stage generative adversarial network with a dual-attention mechanism (Self-Attention + Cross-Attention) and BERT text encoding to generate realistic 256×256 images from textual descriptions, trained on the CUB-200 and Oxford-102 datasets.

DDRM with Vision Transformer — Image Restoration Team

- Combined Denoising Diffusion Restoration Models (DDRM) with Vision Transformers (ViT) to reconstruct high-quality images from degraded inputs, improving detailed feature extraction and robust recovery across diverse scenarios.

YOLOv8 + ViT — Automated Image Annotation Team

- Integrated YOLOv8 real-time object detection with Vision Transformers to deliver accurate, efficient, and scalable image annotation across varied datasets.

ECG-NAT — ECG Neural Attention Transformer Team

- Built an attention-based model to enhance ECG signal analysis, focusing on key cardiac features to improve diagnostic accuracy and detect subtle anomalies.

Smart Home Automation on Raspberry Pi BSc Project

- Developed a cost-effective home automation system on the Raspberry Pi to control smart devices (lighting, security) with remote management and a user-friendly interface, leveraging the board's processing and connectivity capabilities.

TOOLS & TECHNOLOGIES

AI Tools & Platforms: ChatGPT, Claude, Gemini, GitHub Copilot, Hugging Face, Ollama, OpenAI API, LangChain, Midjourney, DALL·E

Software & IDEs: Visual Studio, Jupyter Notebook, Anaconda, Arduino IDE, Overleaf, TexStudio, Draw.io

Data & Design: Power BI, Excel, Word, PowerPoint, Adobe Photoshop, Canva, Camtasia

LANGUAGES

Kurdish	Native
Persian	Fluent
English	Professional Working Proficiency
Arabic	Working Proficiency